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FIN302

Finance and Markets

Lecture 12: Portfolio Insurance and Option Hedging

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Lecture 12-Portfolio Insurance and Option Hedging

Questions and Learning Outcomes

- How to insure portfolio value using index options?
 - Portfolio Insurance
- How to measure option risk?
 - Greek Letters
- How to hedge options positions?
 - Naked and covered positions
 - Stop-loss strategy
- How to make hedging more effective?
 - Dynamic Delta-hedging



Lecture 12-Portfolio Insurance and Option Hedging

- Readings:
 - Hull (2022), Chapter 19
- Tutorial Problems from Hull:
 - 19.11, 30, 31

Portfolio Insurance

- *Portfolio insurance* refers to the strategy of purchasing puts on an equity index to insure a stock portfolio against a possible downturn in the market.
- **Example:** How many three-month puts on the S&P 500 index are required to prevent the value of a \$500,000 portfolio from falling below \$450,000 after three months? What should their strike price be if the current value of the index is 1,000? How does the strategy perform if the index value drops to 880 after three months? Assume $\beta = 1$.

- Since each contract on the S&P 500 index is for \$100 times the index value,

$$\frac{500,000}{1000 \times \$100} = 5$$

three-month puts on the S&P 500 index should be purchased.

Portfolio Insurance

■ Example (continued):

- The strike price of these contracts should be

$$r_P = E(r_M) = -10\%$$

$$K = 1000 * (1 - 0.10) = 900.$$

- If the index value drops to 880, then the expected value of the portfolio will be

$$\frac{880}{1,000} \times 500,000 = \$440,000.$$

- At the same time, the payoff from the put options will be

$$5 \times (900 - 880) \times \$100 = \$10,000.$$

- The expected value of the total hedged portfolio will be

$$440,000 + 10,000 = \$450,000,$$

as required.

- This strategy involves an insurance premium equal to the premium of the put options required to be purchased.

Portfolio Insurance

- If the beta of the portfolio is not 1.0, then the number of contracts must be modified, as well as their strike price.
- **Example:** Suppose a \$500,000 portfolio has a beta of 2.0, while the risk-free interest rate and the dividend yield on the portfolio and the S&P 500 index are 12% per annum and 4% per annum, respectively (both expressed as simple rates). How many three-month puts on the S&P 500 index are required to prevent its value from falling below \$450,000 after three months? What should their strike price be if the current value of the index is 1,000? How does the strategy perform if the index value increases to 1,040 after three months?

- In this case

$$2 \times \frac{500,000}{1000 \times \$100} = 10$$

three-month puts on the S&P 500 index should be purchased.

Portfolio Insurance

■ Example (continued):

- To calculate their strike price, observe that we wish to prevent the total return on the portfolio from being less than

$$r_P = \frac{450,000}{500,000} - 1 + 0.04 \times \frac{3}{12} = -9\%.$$

- According the Capital Asset Pricing Model (CAPM), this corresponds with an expected total index return of

$$\begin{aligned} E(r_M) &= r_F + \frac{r_P - r_F}{\beta_P} \\ &= 0.12 \times \frac{3}{12} + \frac{-0.09 - 0.12 \times \frac{3}{12}}{2} \\ &= -3\%. \end{aligned}$$

- The corresponding return on the index value is thus

$$-0.03 - 0.04 \times \frac{3}{12} = -4\%.$$

Portfolio Insurance

■ Example (continued):

- The desired strike price for the options is therefore

$$1,000 \times (1 - 0.04) = 960.$$

- Next, an index value of 1,040 after three months corresponds with a total index return of

$$r_M = \frac{1,040}{1,000} - 1 + 0.04 \times \frac{3}{12} = 5\%.$$

- According to the CAPM, the expected total return on the portfolio is thus

$$\begin{aligned} E(r_P) &= r_F + \beta_P(r_M - r_F) \\ &= 0.12 \times \frac{3}{12} + 2 \times \left(0.05 - 0.12 \times \frac{3}{12}\right) \\ &= 7\%. \end{aligned}$$

Portfolio Insurance

■ Example (continued):

- The expected return on the value of the portfolio (assuming dividends are not reinvested) is therefore $0.07 - 0.04 \times \frac{3}{12} = 6\%$.
- The corresponding expected portfolio value is $500,000 \times (1 + 0.06) = \$530,000$.
- This is also the total expected value of the hedged portfolio, since the puts expire out-the-money.
- The Table below summarizes the calculations presented above.

Value of index in three months:	1,040
Return from change in index:	40/1,000, or 4% per three months
Dividends from index:	$0.25 \times 4 = 1\%$ per three months
Total return from index:	$4 + 1 = 5\%$ per three months
Risk-free interest rate:	$0.25 \times 12 = 3\%$ per three months
Excess return from index over risk-free interest rate:	$5 - 3 = 2\%$ per three months
Expected excess return from portfolio over risk-free interest rate:	$2 \times 2 = 4\%$ per three months
Expected return from portfolio:	$3 + 4 = 7\%$ per three months
Dividends from portfolio:	$0.25 \times 4 = 1\%$ per three months
Expected increase in value of portfolio:	$7 - 1 = 6\%$ per three months
Expected value of portfolio:	$\$500,000 \times 1.06 = \$530,000$

Greeks: Delta Δ

- Δ : the delta of an option measures the **sensitivity of its price to small changes in the price of its underlying asset**;
- Or the sensitivity of the price of a portfolio of derivatives with a common underlying asset to a change in the price of the underlying asset:

$$\Delta = \frac{\partial \Pi}{\partial S_0}$$

- Thus for options on a non-dividend-paying stock:

$$\Delta_{call} = \frac{\partial c}{\partial S_0} = N(d_1) > 0, \quad \Delta_{put} = \frac{\partial p}{\partial S_0} = N(d_1) - 1 < 0$$

Delta Hedging

Example: Consider a 20 week European call of a non-dividend-paying stock, with $K = \$50$ per share and $S_0 = 49$ per share, its volatility was 20% per annum, and the continuously compounded risk-free interest rate was 5% p.a.

$$d_1 = \frac{\ln \frac{49}{50} + \left(0.1 + \frac{0.2^2}{2}\right) \frac{20}{52}}{0.2 \sqrt{\frac{20}{52}}} = 0.0542$$

- The delta of the call $\Delta_{call} = N(d_1) = 0.522$.
- This implies that **if the stock price increases by \$1, the option value increases by $0.522 \times \$1 = \0.522 .**
- To hedge a long (short) position in this call option, one must sell (purchase) 0.522 shares at initiation.

Greeks: Gamma Γ

- The gamma of a portfolio of derivatives with a common underlying asset measures the **sensitivity of its delta to a change in the price of the underlying asset**:

$$\Gamma = \frac{\partial \Delta_{\Pi}}{\partial S_0} = \frac{\partial^2 \Pi}{\partial^2 S_0}$$

- The gamma of a European call or a European put on a non-dividend-paying stock is given by

$$\Gamma = \frac{N'(d_1)}{S_0 \sigma \sqrt{T}}, \text{ with } N'(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

Greeks: Gamma Γ

- **Example:** Consider again a 20-week European call with $K = \$50$, $S_0 = \$49$; $r = 5\%$, and $\sigma = 20\%$.

$$\Gamma = \frac{N'(d_1)}{S_0 \sigma \sqrt{T}} \approx \frac{N'(0.0542)}{49 \times 0.2 \times \sqrt{\frac{20}{52}}} \approx 0.066.$$

- This implies that ***if the stock price increases by \$1, the option's delta increases by Gamma in asset = 0.066 x \$1 = 0.066.*** Then the ***new Delta = 0.522 + 0.066 = 0.588.***

Hedging Gamma and Delta

- The gamma of a portfolio can only be altered by a position in an instrument, such as an option, whose gamma is non-zero.
 - The underlying asset has a gamma of zero, thus it cannot be used to change the gamma of the portfolio.
- If w_T option contracts, each with a gamma of Γ_T , is added to a portfolio of derivatives on the same underlying asset, whose gamma is Γ , then the resulting portfolio gamma will be $w_T \Gamma_T + \Gamma$. Thus to make the portfolio gamma-neutral,

$$w_T \Gamma_T + \Gamma = 0, \quad w_T = -\frac{\Gamma}{\Gamma_T}$$

- Since this trade also changes the portfolio delta, due to the fact that the option has a non-zero delta, an additional position in the underlying asset is required to make the portfolio delta neutral.

Hedging Gamma and Delta

Example: Suppose a delta-neutral portfolio of derivatives on a certain asset has a gamma of -3,000, while an traded option on the same underlying asset has a delta of 0.62 and a gamma of 1.5.

- The portfolio can be rendered gamma-neutral by purchasing

$$w_T = -\frac{\Gamma}{\Gamma_T} = -\frac{-3000}{1.5} = 2000 \text{ options.}$$

- After doing so, the delta of the portfolio will be $2000 \times 0.62 = 1,240$.
- So delta-neutrality is restored by selling 1,240 units of the underlying asset.

Hedging Gamma and Delta

- Delta-neutrality provides protection **against relatively small changes in the price** of the underlying asset between rebalancing.
- Gamma-neutrality provides protection **against larger changes in the price** of the underlying asset.

Vega: V

- The vega of a portfolio of derivatives with a common underlying asset measures the sensitivity of its value to a change in the volatility of the price of the underlying asset:

$$\mathcal{V} = \frac{\partial \Pi}{\partial \sigma}$$

- The vega of a European call or a European put on a non-dividend-paying stock is given by

$$\mathcal{V} = S_0 \sqrt{T} N'(d_1).$$

Vega: V

- Example: Consider again a 20-week European call with $K = \$50$, $S_0 = \$49$; $r = 5\%$, and $\sigma = 20\%$. Recall that $d_1 = 0.0542$.
 - The vega of the option is therefore

$$V = S_0 \sqrt{T} N'(d_1) \approx 49 \times \sqrt{\frac{20}{52}} \times N'(0.0542) \approx 12.1$$

- In other words, ***a 1% increase in the volatility of the stock price generates an increase in the option value of approximately $12.1 \times 0.01 = \$0.121$.***

Hedge Vega and Gamma

Example: Suppose a delta-neutral portfolio of derivatives on a certain asset has a gamma of -5,000 and a vega of -8,000, while the following two contracts on the same underlying asset are available for trade:

	Delta	Gamma	Vega
Option 1	0.6	0.5	2.0
Option 2	0.5	0.8	1.2

- The quantities w_1 and w_2 of Option 1 and Option 2, respectively, that must be added to the portfolio to make it gamma- and vega-neutral are determined by
$$\begin{aligned}-5,000 + 0.5w_1 + 0.8w_2 &= 0 \\ -8,000 + 2.0w_1 + 1.2w_2 &= 0.\end{aligned}$$
- Solving this system of equations yields $w_1 = 400$ and $w_2 = 6,000$.

Hedge Delta, Vega, and Gamma

Example:

- So the portfolio will be gamma- and vega-neutral after purchasing 400 contracts of Option 1 and 6,000 contracts of Option 2.
 - However, the delta of the portfolio will then be $400 \times 0.6 + 6,000 \times 0.5 = 3,240$.
 - Consequently, 3,240 units of the underlying asset must be sold to restore the portfolio's delta-neutrality.

The Greek Letters of a Portfolio

- The Greek letters of a portfolio of options written on a common underlying asset can be calculated from the Greek letters of its components.
- For instance, for a portfolio containing N options on a given asset, the portfolio delta is given by

$$\Delta = \sum_{i=1}^N w_i \Delta_i$$

where Δ_i is the delta of the i-th contract, and w_i is the position in that contract.

- The same principle apply for all other Greek letters.
- Note that a position in units of the underlying asset will then render the entire portfolio delta-neutral.

The Greek Letters of a Portfolio

Example: Consider a portfolio containing the following positions in options on a certain stock:

- A long position in 100,000 three-month calls strike at \$55, each with a delta of 0.533
- A short position in 200,000 five-month calls strike at \$56, each with a delta of 0.468
- A short position in 50,000 two-month puts strike at \$56, each with a delta of -0.508
- The delta of the entire portfolio is therefore
$$100,000 \times 0.533 + (-200,000) \times 0.468 \\ + (-50,000) \times (-0.508) = -14,900$$
- Consequently, a long position in 14,900 shares is required to make the portfolio delta-neutral.

Naïve Strategies

- In this lecture we consider the problem of a trader who has written a European option on a non-dividend paying stock, and wishes to hedge his position optimally by trading the stock.
- **Example:** A financial institution sells a 20-week European call on 100,000 shares of a non-dividend-paying stock for \$300,000. The initial stock price is \$49 per share and the option strike is \$50 per share, while the continuously compounded risk-free interest rate is 5% per annum and the volatility of the stock price is 20% per annum.
 - The Black-Scholes price of the option is \$240,000.
 - Theoretically, the financial institution should therefore be able to realize a profit of \$60,000.
 - In order to do so, it must be able to manage the risks arising from its position.

Naïve Strategies: Naked Positions

- **Example (continued):** Suppose the financial institution in the previous example does nothing to manage the risks associated with its position.

- Its strategy is therefore to maintain a *naked position*.
- The financial institution's profit is

$$\begin{aligned} & 300,000 - 100,000 \times \max\{S_T - 50, 0\} \\ &= \begin{cases} \$300,000 - \$100,000 \times (S_T - \$50) & \text{if } S_T \geq \$50; \\ \$300,000 & \text{if } S_T < \$50. \end{cases} \end{aligned}$$

- So it does well if the option expires out-the-money.
 - However, if the stock price is say \$60 when the option matures, then the financial institution makes a considerable loss of \$700,000.
- We may conclude that a naked short position in the call is simply too risky.

Naïve Strategies: Covered Positions

- **Example (continued):** Suppose the financial institution in the previous example decides to purchase 100,000 shares of stock as soon as the option is sold.

- Its strategy is therefore to adopt a *covered position*.
- The financial institution's profit is

$$\begin{aligned} & 300,000 - 100,000 \times \max\{S_T - 50, 0\} \\ & \quad + 100,000 \times (S_T - 49) \\ & = \begin{cases} \$400,000 & \text{if } S_T \geq \$50; \\ \$300,000 + 100,000 \times (S_T - \$49) & \text{if } S_T < \$50. \end{cases} \end{aligned}$$

- So it does well if the option expires in-the-money.
 - However, if the stock price is say \$40 at maturity, then the financial institution loses \$600,000.
- We may conclude that a covered short position in the call is also too risky. From put-call parity we know that a covered call has the same risks as a naked put.

Stop-Loss Strategy

- As an alternative to both the naked and covered positions, a trader who sells a call may attempt to implement the following *stop-loss strategy*:
 - If the stock price is less than the strike price, then maintain a naked short call.
 - If the stock price is greater than the strike price, then maintain a covered call.
- To do this, the trader must attempt to purchase one share of stock as soon as the stock price exceeds the strike, and he must attempt to sell one share of stock as soon as the stock price falls below the strike price.
- This will guarantee that he holds precisely one share of stock to deliver to his counter-party if the option expires in-the-money, while he holds no stock if the option expires out-the-money.
- So the stop-loss strategy is essentially a combination of the naked and covered positions.

Stop-Loss Strategy

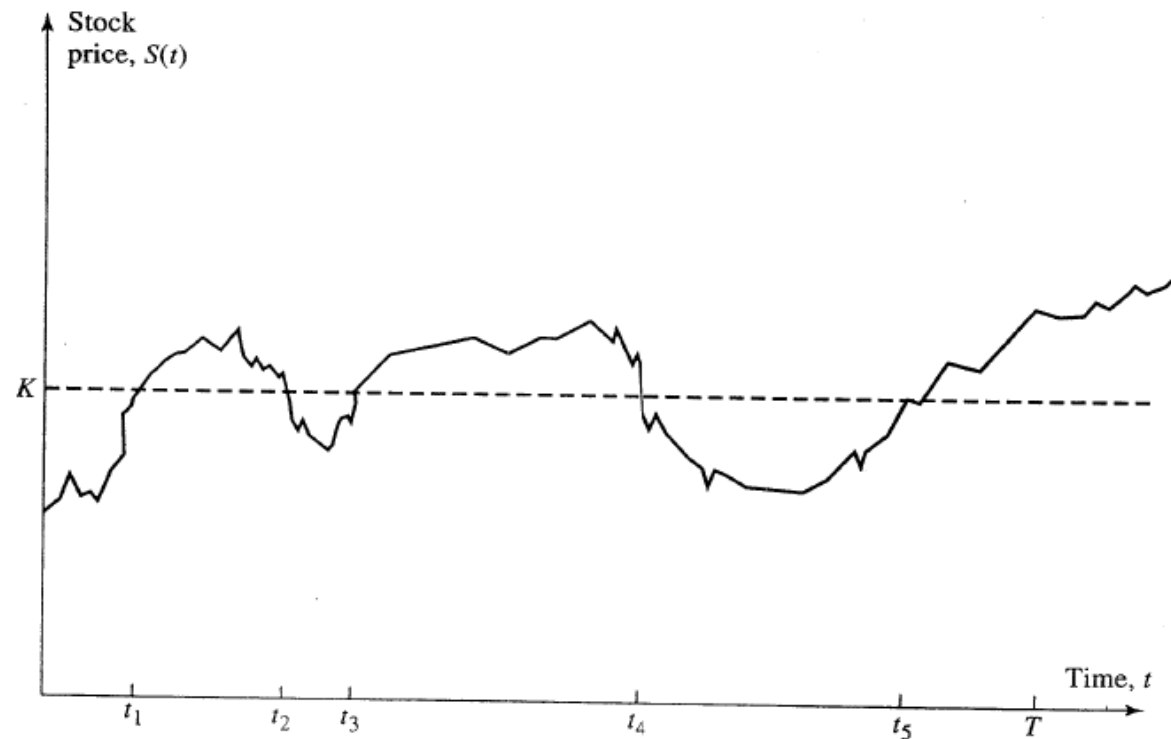


Figure 1: A stop-loss strategy.

The total cost of the strategy is

$$S_{t_1} - S_{t_2} + S_{t_3} - S_{t_4} + S_{t_5} - K = K - K + K - K + K - K = 0.$$

Stop-Loss Strategy

- In general, we may analyze the stop-loss strategy as follows:
 - The initial cost of setting-up the hedge is S_0 if $S_0 > K$, and zero otherwise. All subsequent purchases and sales are made when the stock price is K .
 - So the total cost of writing and hedging the call is simply its intrinsic value at initiation: $Q := \max\{S_0 - K, 0\}$.
- The above analysis is incorrect for two reasons:
 - It ignores the time-value of money, since the hedger's cash-flows are not discounted.
 - It is not possible to purchase and sell stocks for K precisely.
- The first point above is not important, since we could easily incorporate the effect of discounting in our analysis.

Stop-Loss Strategy

- The second objection to our analysis of the stop-loss strategy is critical and may be justified by appealing to the *efficient market hypothesis*:
 - If the market is efficient, then the trader cannot know whether the stock price will go up or down after it hits the level K .
- Thus in practice, the trader may decide to purchase the stock when its price hits the level $K + \varepsilon$, and to sell it when its price hits the level $K - \varepsilon$, for some small $\varepsilon > 0$.
- In that case, each purchase and subsequent sale of the stock costs the trader 2ε (ignoring transaction costs).
- In implementing the stop-loss strategy, the trader therefore faces a tradeoff between frequent smaller losses and less frequent larger ones.
- In practice, the stop-loss strategy does not work very well.

Delta Hedging

- **Example:** A trader has sold 20 call option contracts (20×100 options) on a non-dividend-paying stock. The current stock price is \$100 and call price is \$10, and the delta of the call option is 0.60.
 - Since the call options are written on $20 \times 100 = 2,000$ shares, the trader can hedge his position by purchasing $0.6 \times 2,000 = 1,200$ shares of the underlying stock.
 - $\Delta = 0.60$ implies that for each unit change in the stock price, the value of the option will change by Δ .
 - ▶ If the stock price increases by \$1, then he will lose approximately $0.6 \times 2,000 = \$1,200$ from his short option position, but his long stock position will gain $1 \times 1,200 = \$1,200$.
 - ▶ If the stock price decreases by \$1, then he will gain approximately $0.6 \times 2,000 = \$1,200$ from his short option position, but his long stock position will lose $1 \times 1,200 = \$1,200$.

Dynamic Delta Hedging

- The delta of the trader's option position in the previous example was $-0.6 \times 20 \times 100 = -1,200$.
- However, since one share of stock has a delta of 1, his long stock position had a delta of $1 \times 1,200 = 1,200$.
- The cumulative delta of his option and stock positions was thus zero, which meant that his total portfolio was immune to small changes in the price of the underlying stock.
- We say that his total stock and option position was thus *delta-neutral*.
- Note that since the delta of an option is itself a function of the price of the underlying asset, a portfolio that is initially delta-neutral will not remain delta-neutral after the price of the underlying asset changes.
- To maintain delta-neutrality, the hedge must therefore be *rebalanced* periodically.

Dynamic Delta Hedging

- The delta-hedging procedure described above is an example of a *dynamic hedging strategy*.
- By contrast, in the case of a *static hedging strategy* the hedge is never adjusted after being set-up initially.
- **Example:** With respect to the previous example, suppose that the stock price increases from \$100 to \$110 after three days, while the delta of the call option increases from 0.6 to 0.65.
 - The delta of the trader's short option position is now $-0.65 \times 20 \times 100 = -1,300$.
 - However, the delta of his long stock position is still 1,200.
 - To maintain delta-neutrality, he must therefore purchase an additional 100 shares of stock.

Dynamic Delta Hedging

- Tables 1–2 provide two examples of the dynamic delta-hedging procedure, with weekly rebalancing, applied to the short option position from the previous example.
- In both cases the financial institution must borrow $52,200 \times 49 = \$2,557,800$ in order to make the portfolio delta-neutral at inception.
- Over the first week the interest on this loan amounts to

$$2,557,800 \times \left(e^{0.05 \times \frac{1}{52}} - 1 \right) = \$2,460.606 \approx \$2,500.$$

- In Table 1 the stock price declines to \$48.12 after the first week, and the delta of the option declines to 0.458.
- The financial institution must therefore sell $52,200 - 0.458 \times 100,000 = 6,400$ shares, yielding a cash income of $6,400 \times 48.12 = \$307,968 \approx \$308,000$.

Dynamic Delta Hedging

- The balance of the loan then becomes $2,557,800 + 2,500 - 308,000 = \$2,252,300$, resulting in an interest payment for the second week of

$$2,252,300 \times \left(e^{0.05 \times \frac{1}{52}} - 1 \right) = \$2,166.715 \approx \$2,200.$$

- After the second week the stock price declines again to \$47.37, which results in an associated decline in the option delta, and the hedging process is repeated.
- At the end of the twentieth week the option expires in-the-money in Table 1, and the financial institution ends up with a long position in 100,000 shares of stock and an outstanding loan balance of \$5,263,300.
- The financial institution sells the shares for \$5,000,000 when its counter-party exercises the call, yielding a total cost of hedging of $5,263,300 - 5,000,000 = \$263,300$.

Dynamic Delta Hedging

<i>Week</i>	<i>Stock price</i>	<i>Delta</i>	<i>Shares purchased</i>	<i>Cost of shares purchased (\$000)</i>	<i>Cumulative cost including interest (\$000)</i>	<i>Interest cost (\$000)</i>
0	49.00	0.522	52,200	2,557.8	2,557.8	2.5
1	48.12	0.458	(6,400)	(308.0)	2,252.3	2.2
2	47.37	0.400	(5,800)	(274.7)	1,979.8	1.9
3	50.25	0.596	19,600	984.9	2,966.6	2.9
4	51.75	0.693	9,700	502.0	3,471.5	3.3
5	53.12	0.774	8,100	430.3	3,905.1	3.8
6	53.00	0.771	(300)	(15.9)	3,893.0	3.7
7	51.87	0.706	(6,500)	(337.2)	3,559.5	3.4
8	51.38	0.674	(3,200)	(164.4)	3,398.5	3.3
9	53.00	0.787	11,300	598.9	4,000.7	3.8
10	49.88	0.550	(23,700)	(1,182.2)	2,822.3	2.7
11	48.50	0.413	(13,700)	(664.4)	2,160.6	2.1
12	49.88	0.542	12,900	643.5	2,806.2	2.7
13	50.37	0.591	4,900	246.8	3,055.7	2.9
14	52.13	0.768	17,700	922.7	3,981.3	3.8
15	51.88	0.759	(900)	(46.7)	3,938.4	3.8
16	52.87	0.865	10,600	560.4	4,502.6	4.3
17	54.87	0.978	11,300	620.0	5,126.9	4.9
18	54.62	0.990	1,200	65.5	5,197.3	5.0
19	55.87	1.000	1,000	55.9	5,258.2	5.1
20	57.25	1.000	0	0.0	5,263.3	

Table 1: Simulated delta-hedging, with in-the-money expiry.

Dynamic Delta Hedging

- In the scenario examined in Table 2, the option expires out-the-money, and the financial institution has no position in the underlying stock at the end of the twentieth week.
- In this case the total cost of hedging is simply the final balance of the loan, which amounts to \$256,600.
- In both of the scenarios considered in Tables 1–2 we see that the discounted total cost of hedging is comparable with the initial Black-Scholes premium of \$240,000.
- In particular, by implementing a delta-hedging strategy the financial institution is able to secure the bulk of its theoretical \$60,000 profit, irrespective of the behaviour of the stock price over the life of the option.

Dynamic Delta Hedging

<i>Week</i>	<i>Stock price</i>	<i>Delta</i>	<i>Shares purchased</i>	<i>Cost of shares purchased (\$000)</i>	<i>Cumulative cost including interest (\$000)</i>	<i>Interest cost (\$000)</i>
0	49.00	0.522	52,200	2,557.8	2,557.8	2.5
1	49.75	0.568	4,600	228.9	2,789.2	2.7
2	52.00	0.705	13,700	712.4	3,504.3	3.4
3	50.00	0.579	(12,600)	(630.0)	2,877.7	2.8
4	48.38	0.459	(12,000)	(580.6)	2,299.9	2.2
5	48.25	0.443	(1,600)	(77.2)	2,224.9	2.1
6	48.75	0.475	3,200	156.0	2,383.0	2.3
7	49.63	0.540	6,500	322.6	2,707.9	2.6
8	48.25	0.420	(12,000)	(579.0)	2,131.5	2.1
9	48.25	0.410	(1,000)	(48.2)	2,085.4	2.0
10	51.12	0.658	24,800	1,267.8	3,355.2	3.2
11	51.50	0.692	3,400	175.1	3,533.5	3.4
12	49.88	0.542	(15,000)	(748.2)	2,788.7	2.7
13	49.88	0.538	(400)	(20.0)	2,771.4	2.7
14	48.75	0.400	(13,800)	(672.7)	2,101.4	2.0
15	47.50	0.236	(16,400)	(779.0)	1,324.4	1.3
16	48.00	0.261	2,500	120.0	1,445.7	1.4
17	46.25	0.062	(19,900)	(920.4)	526.7	0.5
18	48.13	0.183	12,100	582.4	1,109.6	1.1
19	46.63	0.007	(17,600)	(820.7)	290.0	0.3
20	48.12	0.000	(700)	(33.7)	256.6	

Table 2: Simulated delta-hedging, with out-the-money expiry.

Dynamic Delta Hedging

- Table 3 presents the results of a simulation exercise for delta-hedging.
- The hedge performance improves steadily as the length of the time-intervals between rebalancing converges to zero.
- This implies that, irrespective of the path taken by the price of the underlying asset, the present-value of the total cost of hedging is simply the Black-Scholes premium when delta-hedging is used in the continuous-time limit.

Time between hedge rebalancing (weeks):	5	4	2	1	0.5	0.25
Performance measure:	0.43	0.39	0.26	0.19	0.14	0.09



Dynamic Delta Hedging

- The delta risk of an options portfolio can be hedged, in other words, a delta neutral portfolio ($\Delta_P = 0$) can be created, by taking positions in the underlying asset or in a futures contract
- If H_F is the position in futures and H_A is the position in the underlying asset, then $H_F = e^{-(r-q)T^*} H_A$ where T^* is the maturity of the futures contract



Appendix: Theta Θ

- The theta of a portfolio of derivatives with a common underlying asset measures the **sensitivity of its price to the passage of time**:

$$\Theta = \frac{\partial \Pi}{\partial T}$$

- Recall that the value of an option may be decomposed into its
- intrinsic value and its time-value.
 - As the option approaches maturity its time-value gradually dissipates, leaving only its intrinsic value.
 - For this reason the theta of an option is often referred to as its time-decay.
- Thus for options on a non-dividend-paying stock:

$$\Theta(\text{call}) = -\frac{S_0 N'(d_1) \sigma}{2\sqrt{T}} - rKe^{-rT} N(d_2)$$

$$\Theta(\text{put}) = -\frac{S_0 N'(d_1) \sigma}{2\sqrt{T}} + rKe^{-rT} N(-d_2)$$

Appendix: Theta Θ

Example: Consider a 20-week European call struck at \$50 on a non-dividend-paying stock whose current price is \$49. The continuously compounded risk-free interest rate is 5% per annum, and the volatility of the stock price is 20% per annum.

$$d_1 = \frac{\ln \frac{S_0}{K} + (r + \frac{1}{2}\sigma^2)T}{\sigma \sqrt{T}} = \frac{\ln \frac{49}{50} + (0.05 + \frac{1}{2} \times 0.2^2) \times \frac{20}{52}}{0.2 \times \sqrt{\frac{20}{52}}} \approx 0.0542$$

$$d_2 = d_1 - \sigma \sqrt{T} \approx -0.0699.$$

$$\begin{aligned}\Theta &= -\frac{S_0 N'(d_1) \sigma}{2 \sqrt{T}} - r K e^{-rT} N(d_2) \\ &\approx -\frac{49 \times N'(0.0542) \times 0.2}{2 \times \sqrt{\frac{20}{52}}} - 0.05 \times 50 \times e^{-0.05 \times \frac{20}{52}} \times N(-0.0699) \\ &\approx -4.31.\end{aligned}$$

So the option value changes by $-4.31 / 365 = -0.0118$ per calendar day and $-4.31 / 252 = -0.0171$ per trading day.



Appendix:

Vega V

- The vega of a portfolio of derivatives with a common underlying asset measures the sensitivity of its value to a change in the volatility of the price of the underlying asset:

$$V = \frac{\partial \Pi}{\partial \sigma}$$

- The vega of a European call or a European put on a non-dividend-paying stock is given by

$$V = S_0 \sqrt{T} N'(d_1).$$

- Example: Consider again a 20-week European call with $K = \$50$, $S_0 = \$49$; $r = 5\%$, and $\sigma = 20\%$. Recall that $d_1 = 0.0542$.

- The gamma of the option is therefore

$$V = S_0 \sqrt{T} N'(d_1) \approx 49 \times \sqrt{\frac{20}{52}} \times N'(0.0542) \approx 12.1$$

- In other words, a 1% increase in the volatility of the stock price generates an increase in the option value of approximately $12.1 \times 0.01 = \$0.121$.



Appendix:

Hedge Vega and Gamma

- The vega of a portfolio can only be altered by a position in an instrument, such as an option, whose vega is non-zero.
 - The underlying asset for a portfolio of derivatives has a vega of zero.
- If w_T option contracts, each with a vega of V_T , is added to a delta-neutral portfolio of derivatives on the same underlying asset, whose vega is denoted by V , then the resulting portfolio vega will be $w_T V_T + V$. The portfolio will be vega-neutral if

$$w_T V_T + V = 0, \quad w_T = -\frac{V}{V_T}$$

- Since this trade also changes the portfolio delta, an additional position in the underlying asset is required to make the portfolio delta-neutral.
- Rendering a portfolio of derivatives with a common underlying asset both gamma- and vega-neutral requires positions in two traded derivatives on the asset. Then again an additional position in the underlying asset is required to restore its delta-neutrality.

Appendix:

Hedge Vega and Gamma

Example: Suppose a delta-neutral portfolio of derivatives on a certain asset has a gamma of -5,000 and a vega of -8,000, while the following two contracts on the same underlying asset are available for trade:

	Delta	Gamma	Vega
Option 1	0.6	0.5	2.0
Option 2	0.5	0.8	1.2

- The quantities w_1 and w_2 of Option 1 and Option 2, respectively, that must be added to the portfolio to make it gamma- and vega-neutral are determined by

$$-5,000 + 0.5w_1 + 0.8w_2 = 0$$

$$-8,000 + 2.0w_1 + 1.2w_2 = 0.$$

- Solving this system of equations yields $w_1 = 400$ and $w_2 = 6,000$.

Appendix:

Hedge Vega and Gamma

Example:

- So the portfolio will be gamma- and vega-neutral after purchasing 400 contracts of Option 1 and 6,000 contracts of Option 2.
 - However, the delta of the portfolio will then be $400 \times 0.6 + 6,000 \times 0.5 = 3,240$.
 - Consequently, 3,240 units of the underlying asset must be sold in order to restore the delta-neutrality of the portfolio.

Appendix:

Rho

- The rho of a portfolio of derivatives with a common underlying asset measures the sensitivity of its value to a change in the risk-free interest rate

$$\text{rho} = \frac{\partial \Pi}{\partial r}$$

- We obtain the following expressions for the rhos of a European call and a European put on a non-dividend-paying stock:

$$\text{rho}(\text{call}) = KTe^{-rT}N(d_2)$$

$$\text{rho}(\text{put}) = -KTe^{-rT}N(-d_2)$$

Appendix: Rho

Example: Consider the previous example of a 20-week European call struck at \$50 on a non-dividend-paying stock whose current price is \$49. The continuously compounded risk-free interest rate is 5% per annum, and the volatility of the stock price is 20% per annum.

- As before, we have $d_2 = -0.0699$.
- The rho of the option is therefore

$$\text{rho} = KTe^{-rT}N(d_2) \approx 50 \times \frac{20}{52} \times e^{-0.05 \times \frac{20}{52}} \times N(-0.0699) \approx 8.91$$

- In other words, a 1% increase in the risk-free interest rate generates an increase in the option value of approximately $8.91 \times 0.01 = \$0.0891$.